

Chapter 1 Introduction to Ethical Hacking

Linux Directory Structure

- **/ (root filesystem):** It is the top-level filesystem directory. It must include every file needed to boot the Linux system before another filesystem is mounted. Every other filesystem is mounted on a well-defined and standard mount point because of the root filesystem directories after the system is started.
- **/bin:** This directory includes user executable files.
- **/etc:** It includes the local system configuration files for the host system.
- **/lib:** It includes shared library files that are needed to start the system.
- **/home:** The home directory storage is available for user files. All users have a subdirectory inside /home.
- **/media:** A place for mounting external removable media devices like USB thumb drives that might be linked to the host.
- **/opt:** It contains optional files like vendor supplied application programs that must be placed here.
- **/root:** It's the home directory for a root user. Keep in mind that it's not the '/' (root) file system.
- **/tmp:** It is a temporary directory used by the OS and several programs for storing temporary files. Also, users may temporarily store files here. Remember that files may be removed without prior notice at any time in this directory.
- **/sbin:** These are system binary files. They are executables utilized for system administration.
- **/usr (unix system resources):** They are read-only and shareable files, including executable libraries and binaries, man files, and several documentation types.
- **/var:** Here, variable data files are saved. It can contain things such as MySQL, log files, other database files, email inboxes, web server data files, and much more.

Linux Commands:

System Info:

- whoami
- id
- hostname
- uname, uname -a, uname -r
- top
- htop
- ifconfig
- ip a
- ping
- date
- service servicename start | stop | status | restart
- history

Navigation:

- pwd
- ls , ls -l, ls -a, ls -la, ls /
- help and man
- cd, cd -, cd ..

Working with Files and Directories:**

- touch
- pluma
- cat
- mkdir
- cp
- mv
- rm
- rmdir
- rm -r

Editing files:

- pluma, mousepad, nano

Finding Files and Directories:

- which
- find (location) (options)
Ex: find / -type f -name hacker.txt
- locate filename

Installing Tools:

- apt-cache search
- sudo apt update
- sudo apt install toolname
- sudo apt remove toolname
- sudo apt purge toolname
- sudo updatedb

File Permissions using chmod:

r - permission to read (4)

w - permission to write (2)

x - permission to execute (1)

Resources

Linuxjourney, linuxcommands cheatsheet, ss64, Linux Basics for hackers